

# Replacing the independent-observation likelihood

## A proposal for the HIKET calibration

Draft, 17 August 2026 — *proposal, not implemented*

### In one paragraph

The calibration currently assumes that every soil-carbon measurement is an independent piece of evidence. It is not: the same plot measured in three different campaigns gives three strongly related numbers, and plots near each other give related numbers too. Because of this, the calibration believes it has roughly twice as much information as it really has, and the excess confidence falls precisely on the quantity that controls the headline results. This document proposes a specific, small, closed-form change that fixes the accounting, explains why the obvious alternative (simply assuming larger measurement error) provably cannot work, and states honestly what the change will and will not achieve.

## 1 What the calibration does now

### 1.1 The model prediction

For each plot  $i$  and each measurement occasion  $j$ , the soil model produces a predicted carbon stock, which we write  $f_{ij}(\theta)$ . Here  $\theta$  is the full vector of parameters being calibrated, and the prediction depends on the plot's litter inputs, its climate, and the year of the measurement. Nothing in this document changes  $f$ ; the proposal concerns only how predictions are compared with observations.

### 1.2 The current comparison

Let  $y_{ij}$  be the measured stock. The calibration currently assumes

$$\log y_{ij} = \log f_{ij}(\theta) + e_{ij}, \quad e_{ij} \text{ independent for every } i \text{ and } j. \quad (1)$$

The comparison is made on the logarithmic scale, which means the error is proportional rather than absolute: an error of “ten per cent” means the same thing on a carbon-poor plot and a carbon-rich one. Each  $e_{ij}$  is drawn from a distribution centred on zero with spread  $s$  (currently 0.80 on the log scale, optionally with heavier tails).

The word doing the damage in Equation 1 is **independent**. In the code this assumption is not a stated hypothesis; it is the plain summation

```
sum(dnorm(log(meta$soc_obs), mean = log(SOC_hat), sd = s, log = TRUE))
```

(`calibration_engine_transient.R`, likelihood block). Adding log-densities together *is* the statement that the observations carry separate, non-overlapping information. With 1205 observations, the calibration therefore behaves as though it has 1205 independent pieces of evidence.

## 2 Why that assumption is wrong here

It is wrong in a way we have measured, not merely suspected.

### Plots persist.

If a plot's carbon stock is higher than the model expects in one campaign, it is higher in the next campaign too. The correlation between the 2006 and 2024 measurements of the *same* plot is 0.66–0.81. The formal summary of this is the *intraclass correlation*, which is simply the share of the total variation that is attributable to persistent differences between plots rather than to fluctuation within a plot. Here it is 0.57–0.71: most of the scatter is a fixed property of the plot.

### Neighbours resemble each other.

Grouping plots by latitude band, the band means vary five to six times more than they would if plots were independent draws from one national population.

### Campaigns differ as blocks.

The three campaigns have different mean offsets and different spreads, so measurements share whatever was peculiar to their campaign.

### A few plots dominate.

The residual distribution has kurtosis close to 7, against 3 for a normal distribution: the tails are far heavier than assumed, so a small number of plots exert disproportionate influence.

Taken together, the effective number of independent observations is roughly **half** the nominal 1205.

### Why this specifically matters

Every one of these mechanisms acts on the *overall level* of soil carbon rather than on differences within a plot. The overall level is exactly what pins down the product of mean residence time and the litter-input multiplier. So the calibration is most overconfident precisely in the direction that determines the results we report. An independent symptom: the posterior standard deviation of the input multiplier is about 0.07 on the log scale, meaning the calibration claims to know it to within roughly seven per cent. That claim is not credible given the correlation structure above, and it is why an externally derived prior, however well founded, currently changes almost nothing.

## 3 Why simply assuming a larger error does not work

The natural first reaction is to keep Equation 1 and increase  $s$ . This provably cannot achieve what is needed, for a reason worth stating carefully.

Increasing  $s$  inflates uncertainty in *every* direction of parameter space by the same factor. But the problem is not that we are uniformly overconfident; we are overconfident *about the level* while remaining appropriately confident about within-plot contrasts. To widen the level enough by inflating  $s$  alone,  $s$  would have to reach roughly 1.8, against a measured residual spread of 0.79. That is internally contradictory: the error model would assert a scatter more than twice what the residuals actually show, destroying the self-consistency argument that justifies the choice of  $s$  in the first place.

### The one-sentence reason correlation is the right tool

Introducing correlation between observations widens uncertainty about the level while leaving the total variance of each individual observation unchanged — and nothing else available to us does that.

## 4 What is proposed

### 4.1 The change

Add a single term to Equation 1:

$$\log y_{ij} = \log f_{ij}(\theta) + u_i + e_{ij}, \quad u_i \sim \mathcal{N}(0, \tau^2), \quad e_{ij} \sim \mathcal{N}(0, s^2). \quad (2)$$

The new quantity  $u_i$  carries no second subscript: there is *one*  $u_i$  per plot, shared by all of that plot’s measurements. It represents whatever is persistently true of that plot and unknown to the model. Because the comparison is on the log scale,  $u_i$  acts multiplicatively: it says “this plot is persistently twenty per cent richer than the model believes.” The parameter  $\tau$  controls how large such plot-level offsets typically are, and  $s$  now describes only the remaining, genuinely occasion-to-occasion variation.

### 4.2 This is not an invented convenience

The interpretation is physical and was identified independently of this proposal. No model in the ensemble contains any description of local soil conditions — texture, drainage, parent material, stoniness beyond the stock correction. Those conditions do not change between 1989 and 2024. A persistent per-plot offset is therefore exactly the form the missing information should take. We are writing down a term for something we already know exists, not adding a free variance to buy comfort.

### 4.3 The plot-level offsets are removed analytically, not estimated

A common objection is that Equation 2 adds hundreds of unknowns, one per plot. It does not, because we never estimate them. We integrate them out — that is, we ask what the observations look like once every possible value of  $u_i$  has been averaged over, weighted by how plausible it is. This is standard and exact, and the result has a closed form.

For a plot with  $n_i$  measurements, collect its log-residuals  $r_{ij} = \log y_{ij} - \log f_{ij}(\theta)$  into a vector. After integrating out  $u_i$ , those residuals are jointly normal with covariance

$$\Sigma_i = s^2 I + \tau^2 \mathbf{1}\mathbf{1}^\top, \quad (3)$$

where  $I$  is the identity matrix and  $\mathbf{1}$  is a column of ones. Every diagonal entry is  $s^2 + \tau^2$  (each observation’s total variance) and every off-diagonal entry is  $\tau^2$  (the covariance any two measurements of the same plot share). This pattern — equal variances, equal covariances — is called *compound symmetry*.

Matrices of this shape can be inverted analytically, so no numerical linear algebra is required. Writing  $n_i$  for the number of observations on plot  $i$ , the contribution of that plot to the log-likelihood is

$$\log L_i = -\frac{1}{2} \left[ \frac{1}{s^2} \left( \sum_j r_{ij}^2 - \frac{\tau^2}{s^2 + n_i \tau^2} \left( \sum_j r_{ij} \right)^2 \right) + (n_i - 1) \log s^2 + \log(s^2 + n_i \tau^2) + n_i \log 2\pi \right]. \quad (4)$$

Everything in Equation 4 is a sum over the plot’s own observations. The cost is proportional to  $n_i$ , identical to the current code. Plots remain independent of one another, so the outer summation over plots is unchanged.

## 4.4 Setting $\tau$

We recommend **fixing**  $\tau$  at the measured intraclass correlation rather than estimating it. Two reasons. First, it adds no free parameter, so the comparison between models is not disturbed — a per-model free variance term would let each model choose its own tolerance for misfit, reintroducing exactly the asymmetry the homogenised prior scheme exists to remove. Second,  $\tau$  is directly measurable from the residuals without reference to the calibration, so fixing it is an external constraint rather than a fitted convenience. The relationship is

$$\rho = \frac{\tau^2}{\tau^2 + s^2}, \quad \text{so} \quad \tau^2 = \frac{\rho}{1 - \rho} s^2, \quad (5)$$

with  $\rho$  the intraclass correlation, measured at 0.57–0.71. A single common value across all six models keeps the intercomparison clean; the sensitivity of the results to  $\rho$  within its measured range should be reported.

## 5 What changes, concretely

|                                  | Before                             | After                                                      |
|----------------------------------|------------------------------------|------------------------------------------------------------|
| Error structure                  | one independent draw per plot-year | one draw per plot-year plus one persistent offset per plot |
| Free parameters                  | unchanged                          | unchanged ( $\tau$ fixed externally)                       |
| Cost per plot                    | proportional to $n_i$              | proportional to $n_i$                                      |
| Code touched                     | —                                  | one block in <code>calibration_engine_transient.R</code>   |
| Variance of a plot's mean        | $s^2/n_i$ , shrinks toward zero    | $s^2/n_i + \tau^2$ , does not                              |
| Confidence in the overall level  | too high                           | appropriately reduced                                      |
| Confidence in within-plot change | appropriate                        | essentially unchanged                                      |

The fifth row is the mechanism in a single line. At present, averaging more measurements of a plot drives the uncertainty in that plot's mean toward zero, so accumulating plots drives uncertainty in the national level toward zero as well. Under the proposal a floor of  $\tau^2$  remains, because additional visits to the same plot cannot tell us anything new about that plot's persistent peculiarity. Differences *within* a plot are unaffected, since the shared offset cancels when one measurement is subtracted from another — which is why the trend is largely preserved while the level is properly widened.

## 6 What this will and will not achieve

### State this before running, not after

The purpose of the change is to make the stated uncertainty honest. It is not guaranteed to move any particular parameter, and it must not be judged by whether it does.

A widened level uncertainty allows externally derived priors to carry more weight, because the relative influence of a prior and a likelihood is set by their relative precisions. Rough arithmetic suggests that accounting for plot-level correlation alone would raise the influence of the input-multiplier prior from about seven per cent to roughly fourteen, and that adding the regional level could bring it near thirty.

### These figures are unreliable and should not be quoted

They come from a marginal, one-parameter-at-a-time approximation. That same style of approximation was tested against a full reweighting of the existing posterior draws and failed badly: once two priors change together, the existing samples provide almost no coverage of the new region — an effective sample size of one draw in a thousand. Direction can be anticipated; magnitude cannot. Only running the calibration settles it.

## 7 Open questions to resolve before implementing

1. **Heavy tails.** The current likelihood optionally uses a Student’s  $t$  distribution to accommodate kurtosis near 7. A  $t$  does not decompose into a shared part plus an independent part the way a normal does, so the two features do not simply stack. Either apply compound symmetry to the normal version, or adopt a multivariate  $t$  per plot, which preserves closed form but changes the meaning of the tail parameter. This must be decided deliberately rather than discovered at runtime.
2. **The regional level.** Plots within a latitude band are also correlated. This can be added as a second, nested shared term, still in closed form, at the cost of the outer summation no longer being over independent units. It is the larger of the two effects on the national level and should probably be included, but it is a bigger change and could follow as a second step.
3. **Comparability of log-likelihoods.** Changing the likelihood changes its normalising constant. Log-likelihoods from before and after are therefore *not* comparable, and neither are any quantities derived from them. Comparisons across the change must use quantities computed after the fact, such as root-mean-square error or the per-draw distributions reported in the appendix on posterior RMSE.
4. **Campaign effects.** A campaign-specific *spread* is safe and could be added. A campaign-specific *mean offset* must not be: the trend we are trying to estimate *is* the difference between campaign levels, so a free offset absorbs the signal. This has already been demonstrated — a fitted offset of  $-0.137$  implied that no accumulation had occurred at all.

## 8 Recommended sequence

1. Measure  $\rho$  once more on the current residuals and fix  $\tau$  from it.
2. Implement Equation 4 for the normal branch only, leaving the Student’s  $t$  branch untouched and selectable, so the change is reversible by a switch.
3. Verify on a short run that the log-likelihood is computed correctly, by checking that setting  $\tau = 0$  reproduces the current values exactly.
4. Run the full calibration; compare using RMSE distributions and parameter locations, never log-likelihood.
5. Only then consider the regional level.

Step three is the one that must not be skipped:  $\tau = 0$  reducing exactly to the present likelihood is the only check that distinguishes a correct implementation from a plausible-looking one.